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SYSTEM AND METHOD FOR POSITIONING A TOW VEHICLE AND A TRAILER

Abstract

A method of operating a tow vehicle and a trailer includes identifying at least one of transient state of operating characteristics or steady state operating characteristics. A transient state based estimation is performed if the transient state operating characteristics are identified to determine a transient state based effective trailer wheelbase and a towbar length and a steady state based estimation is performed if the steady state operating characteristics are identified to determine a steady state based effective trailer wheelbase. An estimated effective trailer wheelbase is determined relative to the tow vehicle based on at least one of the transient state based effective trailer wheelbase and the steady state based effective trailer wheelbase. The tow vehicle and the trailer are maneuvered based on the estimated effective trailer wheelbase.

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Background/Summary

INTRODUCTION

[0001] The present disclosure relates to towing a trailer with a tow vehicle and, more particularly, to determining an effective trailer wheelbase and towbar length for operating the tow vehicle.

[0002] A driver of a tow vehicle can encounter several obstacles when towing a trailer. Towing a trailer can influence the dynamics of the tow vehicle and lead the driver of the tow vehicle to operate the vehicle differently due to the increase in length of the tow vehicle and the trailer combined. Additionally, the trailer can articulate relative to the tow vehicle during certain maneuvers, such as during reversing, which requires a change in inputs to the tow vehicle to reverse when compared to reversing without the trailer.

SUMMARY

[0003] Disclosed herein is a method of operating a tow vehicle with a trailer. The method includes identifying at least one of transient state of operating characteristics or steady state operating characteristics of the tow vehicle. A transient state based estimation is performed if the transient state operating characteristics are identified to determine a transient state based effective trailer wheelbase and a towbar length and a steady state based estimation is performed if the steady state operating characteristics are identified to determine a steady state based effective trailer wheelbase. An estimated effective trailer wheelbase is determined relative to the tow vehicle based on at least one of the transient state based effective trailer wheelbase and the steady state based effective trailer wheelbase. The tow vehicle and the trailer are maneuvered based on the estimated effective trailer wheelbase.

[0004] Another aspect of the disclosure may be where maneuvering the tow vehicle and trailer includes determining a position of the tow vehicle and the trailer relative to a vehicle lane based on sensor data from a plurality of sensors on the tow vehicle. A variation is determined between a current path of the tow vehicle along the vehicle lane and a desired path for the tow vehicle along the vehicle lane. A steering correction is applied to the tow vehicle if the variation exceeds a predetermined threshold.

[0005] Another aspect of the disclosure may be where the transient state operating characteristics includes the tow vehicle having a rate of change in steering angle that exceeded a predetermined threshold.

[0006] Another aspect of the disclosure may be where calculating the towbar length and the transient state based effective trailer wheelbase includes performing an adaptive filter or a recursive least square derivation.

[0007] Another aspect of the disclosure may be where the steady state operating characteristics include the tow vehicle having a rate of change in steering angle below a predetermined threshold.

[0008] Another aspect of the disclosure may be where maneuvering the tow vehicle and the trailer includes applying a steering correction to the tow vehicle when positioning the trailer in a parking spot or positioning the trailer along a desired path when reversing the tow vehicle and the trailer.

[0009] Another aspect of the disclosure may be where the estimated effective trailer wheelbase is determined based on merging the transient state based effective trailer wheelbase with the steady state based effective trailer wheelbase.

[0010] Another aspect of the disclosure may be where the transient state operating characteristics and the steady state operating characteristics are collected when a velocity of the tow vehicle is

within a predetermined velocity range.

[0011] Another aspect of the disclosure may include performing the steady state based estimation to determine the steady state based effective trailer wheelbase is based on the towbar length determined from the transient state based estimation.

[0012] Another aspect of the disclosure may be where the current path is determined based on a set of current vehicle operating parameters and the set of current vehicle operating parameters include a longitudinal velocity of the tow vehicle, a steering angle to the tow vehicle, and a lane position of the tow vehicle.

[0013] Another aspect of the disclosure may be where the desired path for the tow vehicle positions the trailer at least a predetermined distance from an edge of a vehicle lane.

[0014] Disclosed herein is a non-transitory computer-readable storage medium embodying programmed instructions which, when executed by a processor, are operable for performing a method. The method includes identifying at least one of transient state of operating characteristics or steady state operating characteristics of the tow vehicle. A transient state based estimation is performed if the transient state operating characteristics are identified to determine a transient state based effective trailer wheelbase and a towbar length and a steady state based estimation is performed if the steady state operating characteristics are identified to determine a steady state based effective trailer wheelbase. An estimated effective trailer wheelbase is determined relative to the tow vehicle based on at least one of the transient state based effective trailer wheelbase and the steady state based effective trailer wheelbase. The tow vehicle and the trailer are maneuvered based on the estimated effective trailer wheelbase.

[0015] Disclosed herein is a vehicle system. They system includes a vehicle body supported by a first pair of wheels and a second pair of wheels, at least one optical sensor, and a controller in data communication with the at least one optical sensor. The controller is configured to identify at least one of transient state of operating characteristics or steady state operating characteristics of the tow vehicle. The controller is also configured to perform a transient state based estimation if the transient state operating characteristics are identified to determine a transient state based effective trailer wheelbase and a towbar length and perform a steady state based estimation if the steady state operating characteristics are identified to determine a steady state based effective trailer wheelbase. An estimated effective trailer wheelbase is determined by the controller relative to the tow vehicle based on at least one of the transient state based effective trailer wheelbase and the steady state based effective trailer wheelbase. The controller is further configured to maneuver the tow vehicle based on the estimated effective trailer wheelbase.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a schematic diagram of a vehicle system including a vehicle and a trailer.

[0017] FIG. 2 illustrates a flowchart of an example method of positioning a tow vehicle and a trailer within a vehicle lane.

[0018] FIG. 3 illustrates a flowchart of an example method of determining a towbar length and an effective trailer wheelbase.

[0019] FIG. 4 schematically illustrates the tow vehicle and the trailer of FIG. 1 configured for determining a transient-based effective trailer wheelbase.

[0020] FIG. 5 schematically illustrates the tow vehicle and the trailer of FIG. 1 configured for calculating a steady state based effective trailer wheelbase.

[0021] FIG. 6 schematically illustrates positioning the tow vehicle and the trailer FIG. 1 within a vehicle lane.

DETAILED DESCRIPTION

[0022] The present disclosure is susceptible to embodiments in many different forms.

Representative examples of the disclosure are shown in the drawings and described herein in detail as non-limiting examples of the disclosed principles. To that end, elements and limitations described in the Abstract, Introduction, Summary, and Detailed Description sections, but not explicitly outlined in the claims, should not be incorporated into the claims, singly or collectively, by implication, inference, or otherwise.

[0023] For purposes of the present description, unless specifically disclaimed, use of the singular includes the plural and vice versa, the terms “and” and “or” shall be both conjunctive and disjunctive, and the words “including”, “containing”, “comprising”, “having”, and the like shall mean “including without limitation”. Moreover, words of approximation such as “about”, “almost”, “substantially”, “generally”, “approximately”, etc., may be used herein in the sense of “at, near, or nearly at”, or “within 0-5% of”, or “within acceptable manufacturing tolerances”, or logical combinations thereof. As used herein, a component that is “configured to” perform a specified function is capable of performing the specified function without alteration, rather than merely having the potential to perform the specified function after further modification. In other words, the described hardware, when expressly configured to perform the specified function, is specifically selected, created, implemented, utilized, programmed, and/or designed for the purpose of performing the specified function.

[0024] Embodiments of the present disclosure may be described herein in terms of functional and/or logical block components and various processing steps. It should be appreciated that such block components may be realized by a number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of the present disclosure may employ various integrated circuit components, e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with a number of systems and that the systems described herein are merely exemplary embodiments of the present disclosure.

[0025] For the sake of brevity, techniques related to signal processing, data fusion, signaling, control, and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. It should be noted that alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure.

[0026] With reference to FIG. 1, the vehicle **10** generally includes a chassis **12**, a body **14**, front and rear wheels **17**. The body **14** is arranged on the chassis **12** and substantially encloses components of the vehicle **10**. The body **14** and the chassis **12** may jointly form a frame. The wheels **17** are each rotationally coupled to the chassis **12** near a respective corner of the body **14**.

[0027] In various embodiments, the vehicle **10** may be an autonomous vehicle or a semi-autonomous vehicle. The vehicle **10** is, for example, a vehicle that is automatically controlled to carry passengers from one location to another. The vehicle **10** is depicted in the illustrated embodiment as a pickup truck, but it should be appreciated that other vehicles including sport utility vehicles (SUVs), recreational vehicles (RVs), etc., can also be used.

[0028] The vehicle **10** is part of a vehicle system **9**. The vehicle system **9** further includes a trailer **11** attached to the vehicle **10**. The trailer **11** includes one or more trailer axles **13** each having trailer wheels **15** for supporting the trailer **11**.

[0029] As shown, the vehicle **10** generally includes a propulsion system **20**, a transmission system **22**, a steering system **24**, a brake system **26**, a sensor system **28**, an actuator system **30**, at least one data storage device **32**, at least one controller **34**, and a communication system **36**. The propulsion

system **20** may, in various embodiments, include an electric machine such as a traction motor and/or a fuel cell propulsion system. The vehicle **10** further includes a battery (or battery pack) **21** electrically connected to the propulsion system **20**. Accordingly, the battery pack **21** is configured to store electrical energy and to provide electrical energy to the propulsion system **20**. Additionally, the propulsion system **20** may include an internal combustion engine. The transmission system **22** is configured to transmit power from the propulsion system **20** to the vehicle wheels **17** according to selectable speed ratios. According to various embodiments, the transmission system **22** may include a step-ratio automatic transmission, a continuously variable transmission, or other appropriate transmission. The brake system **26** is configured to provide braking torque to the vehicle wheels **17**. The brake system **26** may, in various embodiments, include friction brakes, brake by wire, a regenerative braking system such as an electric machine, and/or other appropriate braking systems. The steering system **24** influences a position of the vehicle wheels **17**. While depicted as including a steering wheel for illustrative purposes, in some embodiments contemplated within the scope of the present disclosure, the steering system **24** may not include a steering wheel. [0030] The vehicle wheels **17** are supported by a front axle **19F** and a rear axle **19R**. The front and rear axles **19F** and **19R** are separated by a distance l . The rear axle **19R** is spaced from a receiver **52**, such as a hitch attachment, by a distance $e_{\text{sub.h}}$ and a trailer connection **56** on a hitch mount **54** is spaced from the receiver **52** by a distance $\Delta e_{\text{sub.h}}$. For purposes of this disclosure, when the trailer **11** includes more than one axle **13**, an effective trailer wheelbase $l_{\text{sub.tr}}$ is used to represent how the trailer **11** maneuvers. In particular, the effective trailer wheelbase $l_{\text{sub.tr}}$ represents an average distance of the axles **13** from the trailer connection **56** or pivot between the trailer **11** and the tow vehicle **10**.

[0031] The sensor system **28** includes one or more sensors **40** (i.e., sensing devices) that sense observable conditions of the exterior environment and/or the interior environment of the vehicle **10**. The sensors **40** are in communication with the controller **34** and may include, but are not limited to, one or more radars, one or more light detection and ranging (lidar) sensors, one or more ground penetrating radar (GPR) sensors, one or more global positioning systems (GPS) devices, one or more cameras (e.g., optical cameras and/or thermal cameras, such as a rear camera and/or a front camera), brake pedal position sensor, accelerator pedal position sensor, steering angle sensor, speed sensor, wheel speed sensor, ride height sensors, steering angle sensor, ultrasonic sensors, one or more inertial measurement units (IMUs), and/or other sensors.

[0032] The sensor system **28** includes one or more Global Positioning System (GPS) transceiver configured to detect and monitor the route data (i.e., route information). The GPS device is configured to communicate with a GPS to locate the position of the vehicle **10** in the globe. The GPS device is in electronic communication with the controller **34**. Because the sensor system **28** provides data to the controller **34**, the sensor system **28** and its sensors **40** are considered sources of information (or simply sources).

[0033] The actuator system **30** includes one or more actuator devices **42** that control one or more vehicle features such as, but not limited to, the propulsion system **20**, the transmission system **22**, the steering system **24**, and the brake system **26**. In various embodiments, the vehicle features can further include interior and/or exterior vehicle features such as, but are not limited to, doors, a trunk, and cabin features such as air, music, lighting, etc. (not numbered). For example, the actuator devices **42** include an accelerator pedal, a brake pedal, etc.

[0034] The controller **34** includes at least one processor **44** and a non-transitory computer-readable storage device or media **46**. The processor **44** can be custom-made or commercially available, a central processing unit (CPU), a graphics processing unit (GPU), an auxiliary processor among several processors associated with the controller **34**, a semiconductor-based microprocessor (in the form of a microchip or chip set), a microprocessor, a combination thereof, or generally a device for executing instructions. The computer-readable storage device or media **46** may include volatile and nonvolatile storage in read-only memory (ROM), random-access memory (RAM), and keep-alive

memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the processor **44** is powered down. The computer-readable storage device or media **46** may be implemented using several other memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or another electric, magnetic, optical, or combination memory devices capable of storing data, some of which represent executable instructions, used by the controller **34** in controlling the vehicle **10**.

[0035] The instructions may include one or more separate programs, each of which comprises an ordered listing of executable instructions for implementing logical functions. The instructions, when executed by the processor **44**, receive and process signals from the sensor system **28**, perform logic, calculations, methods and/or algorithms for automatically controlling the components of the vehicle **10**, and generate control signals to the actuator system **30** to automatically control the components of the vehicle **10** based on the logic, calculations, methods, and/or algorithms.

Although a single controller **34** is shown in FIG. **1**, embodiments of the vehicle **10** may include a number of controllers **34** that communicate over a suitable communication medium or a combination of communication mediums and that cooperate to process the sensor signals, perform logic, calculations, methods, and/or algorithms, and generate control signals to automatically control features of the vehicle **10**.

[0036] The vehicle **10** includes a human-machine interface (HMI) **23**, which may be a center stack screen or a drive information center. The HMI **23** interacts with a user to show information and receives confirmation, activation, cancellation commands, etc. The HMI **23** may be configured as an alarm, such as a speaker to provide a sound, a haptic feedback in a vehicle seat or other object, a visual display, or other device suitable to provide a notification to the vehicle operator of the vehicle **10**. The HMI **23** is in electronic communication with the controller **34** and is configured to receive inputs from a user (e.g., vehicle operator). Accordingly, the controller **34** is configured to receive inputs from the user via the HMI **23**. The HMI **23** includes a display configured to display information to the user (e.g., vehicle operator or passenger) and may include one or more speakers to provide an audible notification to the vehicle operator.

[0037] FIG. **2** illustrates a flow diagram of a method **100** of positioning the tow vehicle **10** and the trailer **11** within a vehicle lane **60** as shown in FIG. **6**. When the tow vehicle **10** is pulling the trailer **11** along the vehicle lane **60**, the trailer **11** travels closer to an edge **62** of the vehicle lane **60** than the tow vehicle **10**. To compensate for the proximity of the trailer **11** to the edge **62** of the vehicle lane **60**, the tow vehicle **10** may need to make wider turns than if the tow vehicle **10** did not have the trailer **11**. Two parameters that contribute to determining a trajectory or path that the trailer **11** follows in the vehicle lane **60** include a distance from the trailer connection **56** to a known location on the tow vehicle **10**, such as the rear axle **19R**, and the other parameter includes the effective trailer wheelbase (l.sub.tr). This disclosure will allow these two parameters to be calculated from operation of the tow vehicle **10** and trailer **11** without additional input from the driver of the tow vehicle **10**.

[0038] In the illustrated example, the tow vehicle **10** includes a receiver-style hitch having a receiver **52** that accepts a hitch mount **54** within the receiver **52**. However, other types of trailer connections can benefit from this disclosure, such as fifth-wheel or goose-neck connections. The hitch mount **54** is generally fixed to the receiver **52** with a pin that extends through both the hitch mount **54** and the receiver **52**. The hitch mount **54** also includes the trailer connection **56** adjacent a distal end for connecting to the trailer **11** and allowing the trailer **11** to pivot relative to the tow vehicle **10**.

[0039] Because the receiver **52** is in a fixed position relative to the tow vehicle **10**, its location relative to the vehicle **10** remains unchanged during use. However, the location of the trailer connection **56** relative to the receiver can vary depending on a length of the hitch mount **54** attached to the receiver **52**. As shown in FIGS. **1** and **4**, the receiver **52** or hitch is located a distance

en from a rear axle **19R** on the tow vehicle **10**. The variable distance between the receiver **52** and the trailer connection **56** is captured by the distance $\Delta e_{\text{sub.h}}$. The distance $\Delta e_{\text{sub.h}}$ can be measured by the driver of the tow vehicle **10** but performing this additional step can be time-consuming and require tools that may not be available to the driver of the tow vehicle **10** when connecting to the trailer **11**. One feature of this disclosure is to reduce the number of steps required by the driver of the tow vehicle **10** to take when changing between hitch mounts **54** and trailer **11**. [0040] The method **100** begins at Block **102** by receiving vehicle operating characteristics that are obtained while the tow vehicle **10** is in motion. In one example, the vehicle operating characteristics can include a longitudinal velocity ($V_{\text{sub.x}}$) of the tow vehicle **10**, a hitch angle (γ) formed between the tow vehicle **10** and the trailer **11**, a hitch angle change rate ($\dot{\gamma}$), and a vehicle yaw rate ($w_{\text{sub.z}}$). These operating characteristics are used to estimate the effective trailer wheelbase and a towbar length ($\Delta e_{\text{sub.h}}$) relative to the receiver **52** such that the location of the trailer connection relative to tow vehicle **10** can be determined. The method **100** estimates the effective trailer wheelbase $\{\text{circumflex over } l\}_{\text{sub.tr}}$ by merging (Block **154**) a calculation of a transient state effective trailer wheelbase (Block **150**) and a calculation of a steady-state effective trailer wheelbase (Block **152**) as shown in the expanded view of Block **102** in FIG. **3**. The transient state effective trailer wheelbase is calculated from transient state operating characteristics and the steady state effective trailer wheelbase is calculated from steady-state operating characteristics as will be discussed in greater detail below.

[0041] In one example, the transient state effective trailer wheelbase $l_{\text{sub.tr}}$ from Block **150** (see FIGS. 2-3) can be calculated based on EQ. 1 below.

$$[00001] \quad l_{\text{tr}} = \frac{-(v_x \sin(\gamma) + w_z \cos(\gamma)(e_h + \dot{e}_h))}{w_z + \dot{\gamma}} \quad \text{EQ. 1}$$

[0042] Because a lateral slip angle of the rear tire on the tow vehicle **10** and the trailer tires is assumed to be small, the following assumptions can be made in EQS. 2-4 to simplify EQ. 1.

$$[00002] \quad v_y^h + v_y^{\text{veh}} \cos(\gamma) + v_x \sin(\gamma) = 0 \quad \text{EQ. 2} \quad v_y^h = l_{\text{tr}} \times \dot{\gamma} \quad \text{EQ. 3}$$

$$v_y^{\text{veh}} = (l_{\text{tr}} \cos(\gamma) + e_h) \times w_z \quad \text{EQ. 4}$$

[0043] In EQ. 2 above, $v_{\text{sub.y.sup.h}}$ is a lateral velocity (in the trailer coordinate system) from the hitch angle rate and the relative velocity of the trailer **11** with respect to the vehicle **10**.

$v_{\text{sub.y.sup.veh}}$ is a lateral velocity (in the vehicle coordinate system) at the trailer axle **13** assuming hitch angle rate is zero signifying that the trailer **11** and tow vehicle **10** are operating as a unified mass. $v_{\text{sub.x}}$ is a longitudinal velocity (in the vehicle coordinate system). Utilizing EQS. 2-4, EQ. 1 can be simplified as shown in FIG. 5 below.

$$[00003] \quad l_{\text{tr}} = \frac{-e_h w_z \cos(\gamma) + v_x \sin(\gamma)}{\dot{\gamma} + w_z} \quad \text{EQ. 5}$$

[0044] This allows EQ. 5 to be rearranged as shown in EQ. 6 below to solve for $l_{\text{sub.tr}}$ and $\Delta e_{\text{sub.h}}$.

$$[00004] \quad -(v_x \sin(\gamma) + w_z \cos(\gamma)(e_h)) = [(w_z + \dot{\gamma}) w_z \cos(\gamma)] \begin{bmatrix} l_{\text{tr}} \\ e_h \end{bmatrix} \quad \text{EQ. 6}$$

[0045] Using a recursive least square derivation (RLS), an estimator is developed to estimate $l_{\text{sub.tr}}$ and $\Delta e_{\text{sub.h}}$ simultaneously where $-(v_{\text{sub.x}} \sin(\gamma) + w_{\text{sub.z}} \cos(\gamma)(e_{\text{sub.h}}))$ is used as a measurement in the RLS and $[(w_{\text{sub.z}} + \{\text{dot over } (\gamma)\}) w_{\text{sub.z}} \cos(\gamma)]$ is used as a regressor vector in the RLS. The RLS can then be solved to determine the transient state effective trailer wheelbase $l_{\text{sub.tr}}$ and the towbar length $\Delta e_{\text{sub.h}}$. Alternatively, an adaptive filter can be used to estimate $l_{\text{sub.tr}}$ and $\Delta e_{\text{sub.h}}$.

[0046] To calculate the steady state effective trailer wheelbase at Block **152**, the method **100** must identify steady state operating characteristics for the tow vehicle **10** and the trailer **11**. In one example, the steady state operating characteristics are obtained while the vehicle **10** is performing a

turn of a constant radius at a low speed. However, these operating characteristics can also be obtained when the tow vehicle **10** is moving at a variable speed while still performing the turn with the constant radius of curvature that results in the formation of a hitch angle γ . In one example, the hitch angle γ is above a predetermined threshold value.

[0047] Furthermore, the steady state operating characteristics can be obtained over a variable length of time if a steady state equilibrium is reached regarding the hitch angle γ as determined by a value of $\{\dot{\gamma}\}$ being less than a predetermined threshold value. In one example, the low speed is less than 18.6 miles per hour (30 kilometers per hour) and greater than 3 miles per hour (4.8 kilometers per hour) and the constant turn includes a variation in input steering angle for the vehicle **10** that is less than a predetermined rate of change for the steering angle. However, the steady state operating characteristics can also be obtained during higher speed maneuvers that do not result in nonlinear vehicle dynamics above a predetermined threshold.

[0048] At Block **152**, EQS. 7-9 are used to determine the steady state effective trailer wheelbase $l_{\text{sub.tr}}$. In EQ. 7 below, the value for e_n is obtained from Block **150** and includes the towbar length $\Delta e_{\text{sub.h}}$ added to the distance between the hitch receiver **52** and the rear axle **19R** shown in FIG. **4**. In EQ. 7 below, the distances are taken with respect to a vehicle-based coordinate system relative to a center of mass M of the vehicle **10**. In the illustrated example, $x_{\text{sub.c}}$ defines a longitudinal distance from a center of mass M of the vehicle **10** to a center C of the turning radius R of the vehicle **10**, $y_{\text{sub.c}}$ defines a lateral distance between the center of mass M to the center of the turning radius R , and $l_{\text{sub.r}}$ defines a wheelbase of the vehicle **10**. These values allow for the calculation of the angle $\gamma_{\text{sub.1}}$. The angle $\gamma_{\text{sub.1}}$ is defined between a line extending perpendicular to a central longitudinal axis of the vehicle **10** and a line defining a radius of curvature $R_{\text{sub.h}}$ of the trailer connection **56**.

$$[00005] \quad \gamma_{\text{sub.1}} = \tan^{-1} \left(\frac{x_c - (l_r + e_h)}{y_c} \right) \quad \text{EQ. 7}$$

[0049] With the calculation for angle $\gamma_{\text{sub.1}}$ obtained as described above, the method **100** then proceeds to calculate the angle $\gamma_{\text{sub.2}}$. The angle $\gamma_{\text{sub.2}}$ is obtained by subtracting a sum of the angles $\gamma_{\text{sub.1}}$ and γ from $\pi/2$ or 90 degrees as shown by EQ. 8 below.

$$[00006] \quad \gamma_{\text{sub.2}} = \pi/2 - (\gamma_{\text{sub.1}} + \gamma) \quad \text{EQ. 8}$$

[0050] With the angle $\gamma_{\text{sub.2}}$ from EQ. 8 above, the method **100** can then calculate a length for the radius of a curvature $R_{\text{sub.h}}$ for the trailer connection **56** with EQ. 9 as shown below.

$$[00007] \quad R_h = \sqrt{y_c^2 + (x_c - (l_r + e_h))^2} \quad \text{EQ. 9}$$

[0051] With the length of the radius of curvature of the hitch $R_{\text{sub.h}}$, the method **100** calculates the steady state effective trailer wheelbase with EQ. 10 shown below.

$$[00008] \quad l_{\text{tr}} = R_h \cos(\gamma_{\text{sub.2}}) \quad \text{EQ. 10}$$

[0052] The method **100** then proceeds to Block **154** to merge the transient state effective trailer wheelbase with the steady state effective trailer wheelbase to generate an estimated effective trailer wheelbase $\{\text{circumflex over } (l)\}_{\text{sub.tr}}$. In one example, the method **100** may merge the two effective trailer wheelbases to generate the estimated effective trailer wheelbase $\{\text{circumflex over } (l)\}_{\text{sub.tr}}$ by taking an average of the two. The estimated effective trailer wheelbase $\{\text{circumflex over } (l)\}_{\text{sub.tr}}$ can also be used to prevent trailer sway through the controller **34** signaling to the trailer **11** to apply trailer brakes based on the estimated effective trailer wheelbase $\{\text{circumflex over } (l)\}_{\text{sub.tr}}$.

[0053] With the merged trailer wheelbase, the method **100** proceeds to Block **106**. At Block **106**, the method **100** utilizes the estimated effective trailer wheelbase $\{\text{circumflex over } (l)\}_{\text{sub.tr}}$ and the hitch angle γ in addition to other vehicle parameters from Block **104**, such as the wheelbase/of the tow vehicle **10**, to determine a position of the vehicle system **9** on a roadway, such as the vehicle lane **60**. By determining the position of the vehicle system **9** in the vehicle lane **60**, the

method **100** can compare a predicted or actual path of the vehicle system **9** with a desired path for the vehicle system **9**. The comparison will yield a difference ΔD that represents a variation between the predicted path and the desired path. In one example, the difference ΔD can be represented as a lateral distance **64** between the predicted path and the desired path. The difference ΔD is then provided to Block **108**.

[0054] At Block **108**, the method **100** determines if the difference ΔD would require the method **100** to apply a steering correction $\Delta\delta_{\text{sub.f}}$ to the tow vehicle **10** to move the vehicle system **9** closer to the desired path. The steering correction $\Delta\delta_{\text{sub.f}}$ can be applied to the tow vehicle **10** through the controller **34** in connection with the actuator system **30** and the steering system **24**. If the difference ΔD is greater than a predetermined threshold, then the method **100** outputs the steering correction $\Delta\delta_{\text{sub.f}}$. In one example, the steering correction $\Delta\delta_{\text{sub.f}}$ is sent to the actuator system **30** to apply a change to the current path or trajectory of the tow vehicle **10** through the steering system **24**. The application of the steering correction $\Delta\delta_{\text{sub.f}}$ will ensure that the tow vehicle **10** and trailer **11** remain within the vehicle lane **60**. The method **100** can also provide an alert if the trailer **11** cannot remain within the vehicle lane **60** by applying the steering correction $\Delta\delta_{\text{sub.f}}$. The method **100** can also be applied to positioning the trailer **10** in a parking space or when operating the tow vehicle **10** and trailer **11** in reverse. The steering correction $\Delta\delta_{\text{sub.f}}$ can also be applied when performing other maneuvers, such as during parking or reversing.

[0055] The terms “a” and “an” do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced items. The term “or” means “and/or” unless clearly indicated otherwise by context. Reference throughout the specification to “an aspect”, means that a particular element (e.g., feature, structure, step, or characteristic) described in connection with the aspect is included in at least one aspect described herein, and may or may not be present in other aspects. In addition, it is to be understood that the described elements may be combined in a suitable manner in the various aspects.

[0056] When an element such as a layer, film, region, or substrate is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present.

[0057] Unless specified to the contrary herein, test standards are the most recent standard in effect as of the filing date of this application, or, if priority is claimed, the filing date of the earliest priority application in which the test standard appears.

[0058] Unless defined otherwise, technical, and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which this disclosure belongs.

[0059] While the above disclosure has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made, and equivalents may be substituted for elements thereof without departing from its scope. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the disclosure without departing from the scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiments disclosed but will include embodiments falling within the scope thereof.

Claims

1. A method of operating a tow vehicle with a trailer, the method comprising: identifying at least one of transient state of operating characteristics or steady state operating characteristics; performing a transient state based estimation if the transient state operating characteristics are identified to determine a transient state based effective trailer wheelbase and a towbar length; performing a steady state based estimation if the steady state operating characteristics are identified to determine a steady state based effective trailer wheelbase; determining an estimated effective

trailer wheelbase relative to the tow vehicle based on at least one of the transient state based effective trailer wheelbase and the steady state based effective trailer wheelbase; and maneuvering the tow vehicle and the trailer based on the estimated effective trailer wheelbase.

2. The method of claim 1, wherein maneuvering the tow vehicle and the trailer includes: determining a position of the tow vehicle and the trailer relative to a vehicle lane based on sensor data from a plurality of sensors on the tow vehicle; determining a variation between a current path of the tow vehicle along the vehicle lane and a desired path for the tow vehicle along the vehicle lane; and applying a steering correction to the tow vehicle if the variation exceeds a predetermined threshold.

3. The method of claim 1, wherein the transient state operating characteristics include the tow vehicle having a rate of change in steering angle that exceeded a predetermined threshold.

4. The method of claim 3, wherein calculating the towbar length and the transient state based effective trailer wheelbase includes performing an adaptive filter or a recursive least square derivation.

5. The method of claim 1, wherein the steady state operating characteristics include the tow vehicle having a rate of change in steering angle below a predetermined threshold.

6. The method of claim 5, wherein maneuvering the tow vehicle and the trailer includes applying a steering correction to the tow vehicle when positioning the trailer in a parking spot or positioning the trailer along a desired path when reversing the tow vehicle and the trailer.

7. The method of claim 1, wherein the estimated effective trailer wheelbase is determined based on merging the transient state based effective trailer wheelbase with the steady state based effective trailer wheelbase.

8. The method of claim 7, wherein the transient state operating characteristics and the steady state operating characteristics are collected when a velocity of the tow vehicle is within a predetermined velocity range.

9. The method of claim 7, wherein performing the steady state based estimation to determine the steady state based effective trailer wheelbase is based on the towbar length determined from the transient state based estimation.

10. The method of claim 2, wherein the current path is determined based on a set of current vehicle operating parameters and the set of current vehicle operating parameters include a longitudinal velocity of the tow vehicle, a steering angle to the tow vehicle, and a lane position of the tow vehicle.

11. The method of claim 10, wherein the desired path for the tow vehicle positions the trailer at least a predetermined distance from an edge of a vehicle lane.

12. A non-transitory computer-readable storage medium embodying programmed instructions which, when executed by a processor, are operable for performing a method comprising: identifying at least one of transient state of operating characteristics or steady state operating characteristics of a tow vehicle; performing a transient state based estimation if the transient state operating characteristics are identified to determine a transient state based effective trailer wheelbase and a towbar length; performing a steady state based estimation if the steady state operating characteristics are identified to determine a steady state based effective trailer wheelbase; determining an estimated effective trailer wheelbase relative to the tow vehicle based on at least one of the transient state based effective trailer wheelbase and the steady state based effective trailer wheelbase; and maneuvering the tow vehicle and the trailer based on the estimated effective trailer wheelbase.

13. The computer-readable storage medium of claim 12, wherein the transient state operating characteristics include the tow vehicle having a rate of change in steering angle that exceeded a predetermined threshold.

14. The computer-readable storage medium of claim 13, wherein maneuvering the tow vehicle and the trailer includes: determining a position of the tow vehicle and the trailer relative to a vehicle

lane based on sensor data from a plurality of sensors on the tow vehicle; determining a variation between a current path of the tow vehicle along the vehicle lane and a desired path for the tow vehicle along the vehicle lane; and applying a steering correction to the tow vehicle if the variation exceeds a predetermined threshold.

15. The computer-readable storage medium of claim 14, wherein calculating the towbar length and the transient state based effective trailer wheelbase includes performing an adaptive filter or a recursive least square derivation.

16. The computer-readable storage medium of claim 13, wherein the steady state operating characteristics including the tow vehicle having a rate of change in steering angle below a predetermined threshold.

17. The computer-readable storage medium of claim 16, wherein maneuvering the tow vehicle and the trailer includes applying a steering correction to the tow vehicle when positioning the trailer in a parking spot or positioning the trailer along a desired path when reversing the tow vehicle and the trailer.

18. A vehicle system comprising: a vehicle body supported by a first pair of wheels and a second pair of wheels; at least one optical sensor and a controller in data communication with the at least one optical sensor, wherein the controller is configured to: identify at least one of transient state of operating characteristics or steady state operating characteristics of a tow vehicle; perform a transient state based estimation if the transient state operating characteristics are identified to determine a transient state based effective trailer wheelbase and a towbar length; perform a steady state based estimation if the steady state operating characteristics are identified to determine a steady state based effective trailer wheelbase; determine an estimated effective trailer wheelbase relative to the tow vehicle based on at least one of the transient state based effective trailer wheelbase and the steady state based effective trailer wheelbase; and maneuver the tow vehicle and the trailer based on the estimated effective trailer wheelbase.

19. The vehicle system of claim 18, wherein to maneuver the tow vehicle and the trailer, the controller is configured to: determine a position of the tow vehicle and a trailer relative to a vehicle lane based on sensor data from a plurality of sensors on the tow vehicle; determine a variation between a current path of the tow vehicle along the vehicle lane and a desired path for the tow vehicle along the vehicle lane; and apply a steering correction to the tow vehicle if the variation exceeds a predetermined threshold.

20. The vehicle system of claim 19, wherein the transient state operating characteristics including the tow vehicle having a rate of change in steering angle that exceeded a predetermined threshold and the steady state operating characteristics including the tow vehicle having a rate of change in steering angle below a predetermined threshold.
